

11(a)	quantum / packet / discrete amount of <u>energy</u>	M1
	of electromagnetic radiation	A1
11(b)	$E = hc / \lambda$	C1
	$= (6.63 \times 10^{-34} \times 3.0 \times 10^8) / (540 \times 10^{-9})$	C1
	$= (3.68 \times 10^{-19}) / (1.6 \times 10^{-19})$ $= 2.3 \text{ eV}$	A1
11(c)	Any 4 from: photon absorbed by electron in valence band (1) photon energy > energy of forbidden band (1) electron promoted to conduction band (1) hole left in valence band (1) more charge carriers so lower resistance (1)	B4

Question	Answer	Marks
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